Building a Client-Side Geo/Parquet File Viewer

Built on webassembly, works in browser, and data never leaves your computer

Pukar Bhandari

pukar.bhandari@outlook.com

2026-04-12

Table of contents

Want a quick way to preview what is inside your parquet file? This is a browser-based, zero-backend viewer for Parquet and GeoParquet, built on DuckDB-WASM. Nothing is uploaded; everything runs locally in your tab.

- **Open it three ways**: drop in a local file, paste an HTTP(S) URL, or paste an `s3://` URL for a public bucket. DuckDB fetches only the row groups it needs via range requests.
- **See the schema first**: column names, types and nullability, plus row count, file size, row groups and (for spatial files) the geometry column, CRS and bounding box, all before any row data loads.
- **Query it**: a CodeMirror SQL editor with column autocomplete, or a point-and-click query builder whose generated SQL stays visible and editable.
- **Map it**: GeoParquet and spatial files appear on a MapLibre map, synced with the table in both directions.

> Update
>
> Since the first version it has gained support for Esri JSON geometries (rings, paths and x/y columns), clearer error messages for common failures like CORS, 403/404 and unsupported ZSTD compression (re-save with Snappy or Gzip), and a refreshed theme.

The live app is embedded below.

Open app GitHub

If you work with data and want a quick way to preview the contents of a `.parquet` file, give it a try. The code is on GitHub — issues and PRs welcome.